# Required Capstone Assignment 16.1: Finalizing Your Problem Statement [03:00:00] - Section D

## Learning Outcome Addressed

10. Refine your problem statement to include a research question, expected data sources and structure, expected results and techniques.

***This is a required assignment and counts toward program completion.***

By this point in the program, you will have posed a question to answer in your final capstone submission and met with your Learning Facilitator to refine your question and generate ideas for datasets and techniques. Now it is time to submit the official proposal or problem statement.

On this discussion board, your initial post should contain the following:

- The research question you intend to answer (in one sentence, if possible)
- Your expected data source(s) (as either a link to existing data or a sentence describing where you will source the data from)
- The techniques you expect to use in your analysis
- The expected results
- Why this question is important

That last bullet point is the most important. Remember that your final capstone project will be graded not only on the execution of your plan but also on your ability to translate those results into actionable business intelligence that can be acted on by non-AI/ML personnel. Not only will you have to do the number crunching, but you will also have to distill the results into something that the general public can understand. This last bullet point is your opportunity to explain (in non-AI/ML language) why this question is important. What happens if this question is unanswered? What benefit will your analysis bring?

## Submission Instructions:

Please feel free to read/comment on fellow learners' submissions, but note that responses are not required. This activity is structured as a discussion to allow learners to see each others' questions and possible solutions.

# Capstone Assignment 16.1 – Finalizing Problem Statement:

## Predicting Subscription Renewal Risk and Customer Churn Using Machine Learning

**Required Capstone Assignment 16.1 — Finalizing Your Problem Statement**

## 1. Research Question

Can machine learning identify subscription-based customers at greatest risk of churn and explain the factors associated with that risk, enabling businesses to prioritize retention efforts and improve renewal outcomes?

## 2. Expected Data Sources and Structure

I plan to use the publicly available IBM Telco Customer Churn dataset as a proxy for subscription-based customer relationships. The dataset contains customer-level records with demographics, tenure, contract terms, services, monthly and total charges, payment methods, and a binary churn outcome (Yes/No).

The target variable will be Churn. Predictors will include characteristics available before the churn outcome. Although the data concerns telecommunications customers, the methodology may inform future analyses in SaaS subscription and renewal management. It does not directly contain Salesforce renewal opportunities or verified SaaS renewal outcomes.

## 3. Proposed Analysis and Techniques

I will organize the project using the CRISP-DM methodology: business understanding, data understanding, preparation, modeling, evaluation, and recommendations.

### Data preparation and feature engineering

- Inspect data quality, duplicates, missing values, inconsistent categories, and the balance of churn outcomes.
- Encode categorical variables, scale numerical features where appropriate, and apply transformations within training-only pipelines to avoid data leakage.
- Explore tenure groups, monthly-charge segments, service counts, contract commitment, and payment methods as candidate predictors; assess whether engineered features improve results.

### Exploratory visualizations

- Bar charts of churn rates by contract type, payment method, and internet service.
- Histograms of tenure and monthly charges, and box plots comparing charges for churned and retained customers.
- Correlation heatmaps for numeric relationships, plus confusion matrices and ROC/precision–recall curves for model evaluation.

### Predictive modeling and evaluation

I will compare Logistic Regression, Decision Tree, and Random Forest classifiers using a stratified train/test split, cross-validation, and hyperparameter tuning. Performance will be assessed using recall, precision, F1-score, ROC-AUC, and precision–recall measures. Because missing at-risk customers may be costly, I will examine recall alongside precision and the practical workload created by false positives. Model coefficients or feature-importance analyses will be used to communicate associations in understandable terms, without claiming causation.

## 4. Expected Results

I expect to identify customer segments associated with elevated churn risk, compare the predictive performance of several models, and develop an interpretable customer-risk ranking. Possible associations include tenure, contract length, pricing, payment method, and subscribed services; these are hypotheses to test, not predetermined findings.

The final project will present clear charts, model comparisons, risk-segment summaries, and recommendations that nontechnical business stakeholders can evaluate.

## 5. Why This Question Matters

Subscription businesses depend on recurring customer relationships. When customers leave, organizations lose recurring revenue and may spend additional resources acquiring replacements. Without early warning signals, customer success and renewal teams may identify problems only after a customer has already canceled.

A practical churn-risk model could help teams focus outreach on customers who may need support rather than applying the same intervention to everyone. For example, if short-tenure customers show elevated risk, a business could evaluate stronger onboarding; if certain contract or pricing segments are associated with churn, it could investigate service value or renewal options. These would be proposals for testing, not proven effects of the model.

The intended business value is better prioritization of retention resources, clearer visibility into churn risk, and informed opportunities to protect recurring revenue. Any operational

recommendations should be validated through later-period data or a controlled pilot, with attention to outreach costs, fairness, privacy, and customer consent.

## 6. Planned Next Steps

1. Confirm the exact dataset version and document its fields and limitations.
2. Perform exploratory analysis, preprocessing, model comparison, and interpretation.
3. Translate validated findings into a concise retention playbook for nontechnical decision-makers.
4. Recommend a prospective pilot and monitoring approach before any real-world deployment.